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**Exp No.: 1 Setting up Hyperledger Fabric Development**

Environment

**Aim**

To install and configure the Hyperledger Fabric development environment on a Linux system using Docker, Go, and Fabric binaries, create a blockchain test network, and establish a communication channel between peer organizations.

Software Requirements

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Go (Version 1.20 or later)
- Git
- Curl
- Hyperledger Fabric Samples
- Hyperledger Fabric Docker Images

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB free disk space
- Internet Connection

**Procedure**

**Step 1 :** Install wsl

```
PS C:\Windows\system32> wsl --install
A distribution with the supplied name already exists. Use --name to chose a different name.
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS
PS C:\Windows\system32> wsl --status
Default Distribution: Ubuntu
Default Version: 2
PS C:\Windows\system32> Ubuntu 22.04 LTS
Launches or configures a Linux distribution.

Usage:
    <no args>
        Launches the user's default shell in the user's home directory.

    install [--root]
        Install the distribuiton and do not launch the shell when complete.
          --root
              Do not create a user account and leave the default user set to root.

    run <command line>
        Run the provided command line in the current working directory. If no
        command line is provided, the default shell is launched.

    config [setting [value]]
        Configure settings for this distribution.
        Settings:
          --default-user <username>
              Sets the default user to <username>. This must be an existing user.

    help
        Print usage information and exit.
PS C:\Windows\system32> Ubuntu 22.04 LTS install
Launches or configures a Linux distribution.

Usage:
    <no args>
        Launches the user's default shell in the user's home directory.

    install [--root]
        Install the distribuiton and do not launch the shell when complete.
          --root
              Do not create a user account and leave the default user set to root.
```

**Step 2:** install git,curl,jq,make,gcc,g++

```
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo apt install -y \ git \ curl \ jq \
> ^C
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo apt install -y \
> git \
> curl \
> jq \
> make \
> gcc \
> g++ \
> docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package docker-compose-plugin
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
```

**Step 3**: Install docker

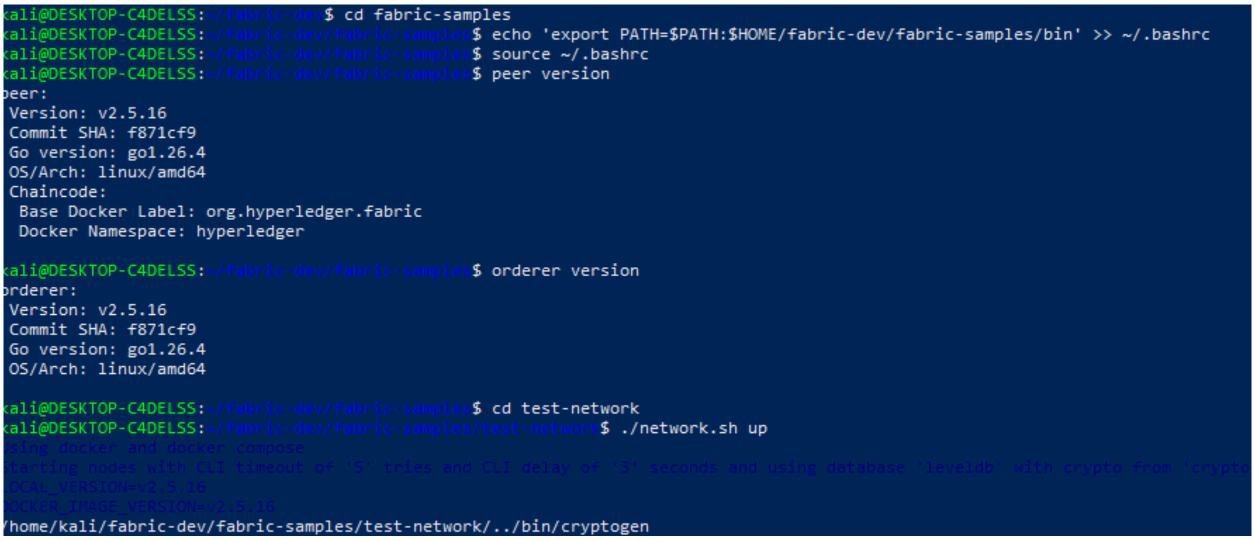
```
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo install -m 0755 -d /etc/apt/keyrings
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
> sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo chmod a+r /etc/apt/keyrings/docker.gpg
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ echo \
> "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
> https://download.docker.com/linux/ubuntu \
> $(. /etc/os-release && echo $VERSION_CODENAME) stable" | \
> sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo apt update
Get:1 https://download.docker.com/linux/ubuntu jammy InRelease [48.5 kB]
Get:2 https://download.docker.com/linux/ubuntu jammy/stable amd64 Packages [83.5 kB]
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:5 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Fetched 132 kB in 1s (94.4 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
131 packages can be upgraded. Run 'apt list --upgradable' to see them.
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  dbus-user-session docker-ce-rootless-extras pigz
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io dbus-user-session docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras
  docker-compose-plugin pigz
```

```
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable docker
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo systemctl start docker
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
     Active: active (running) since Fri 2026-07-24 15:52:47 IST; 1min 6s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 7718 (dockerd)
      Tasks: 10
     Memory: 29.8M
        CPU: 646ms
     CGroup: /system.slice/docker.service
             └─7718 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jul 24 15:52:45 DESKTOP-C4DELSS dockerd[7718]: time="2026-07-24T15:52:45.491251176+05:30" level=info msg="Restori>Jul 24 15:52:45 DESKTOP-C4DELSS dockerd[7718]: time="2026-07-24T15:52:45.6
2949+05:30" level=info msg="Deletin>Jul 24 15:52:45 DESKTOP-C4DELSS dockerd[7718]: time="2026-07-24T15:52:45.700837373+05:30" level=info msg="Deletin>Jul 24 15:52:46 DESKTOP-C4DELSS docker
18]: time="2026-07-24T15:52:46.560613825+05:30" level=info msg="Loading>Jul 24 15:52:46 DESKTOP-C4DELSS dockerd[7718]: time="2026-07-24T15:52:46.572525541+05:30" level=info msg="Docker >Ju
 15:52:46 DESKTOP-C4DELSS dockerd[7718]: time="2026-07-24T15:52:46.572989649+05:30" level=info msg="Initial>Jul 24 15:52:47 DESKTOP-C4DELSS dockerd[7718]: time="2026-07-24T15:52:47.0803833
5:30" level=info msg="Complet>Jul 24 15:52:47 DESKTOP-C4DELSS dockerd[7718]: time="2026-07-24T15:52:47.092387712+05:30" level=info msg="Daemon >Jul 24 15:52:47 DESKTOP-C4DELSS dockerd[7718
ime="2026-07-24T15:52:47.092565454+05:30" level=info msg="API lis>Jul 24 15:52:47 DESKTOP-C4DELSS systemd[1]: Started Docker Application Container Engine.
```

**Step 4:** Install Hyperledger Fabric
**Step 5**:Download Fabric Samples

```
Lines 1-22/22 (END)
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ sudo usermod -aG docker $USER
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ newgrp docker
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ docker --version
Docker version 29.6.2, build dfc4efb
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ docker compose version
Docker Compose version v5.3.1
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:c3cbe1cc1aa588a64951ac6286e0df7b27fe2e6324b1001c619bb358770c0178
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ mkdir -p ~/fabric-dev
kali@DESKTOP-C4DELSS:/mnt/c/Windows/system32$ cd ~/fabric-dev
kali@DESKTOP-C4DELSS:~/fabric-dev$ curl -sSL https://bit.ly/2ysbOFE | bash -s

Clone hyperledger/fabric-samples repo

===> Cloning hyperledger/fabric-samples repo
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15798, done.
```

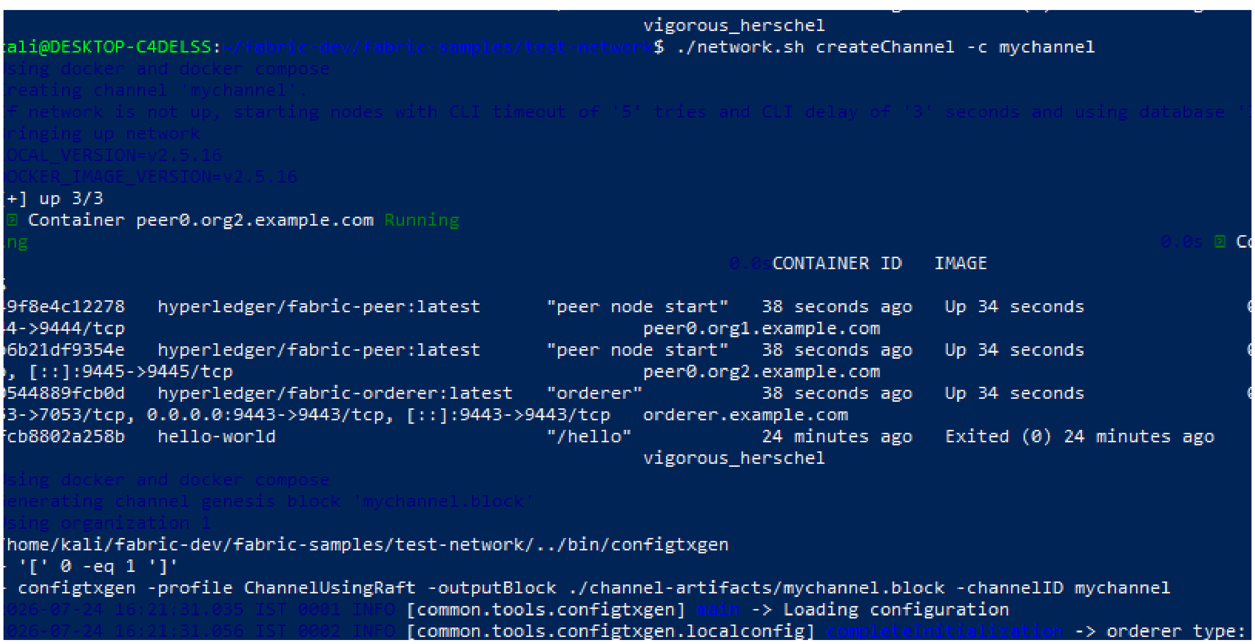
**Step 6:** verify peer and orderer



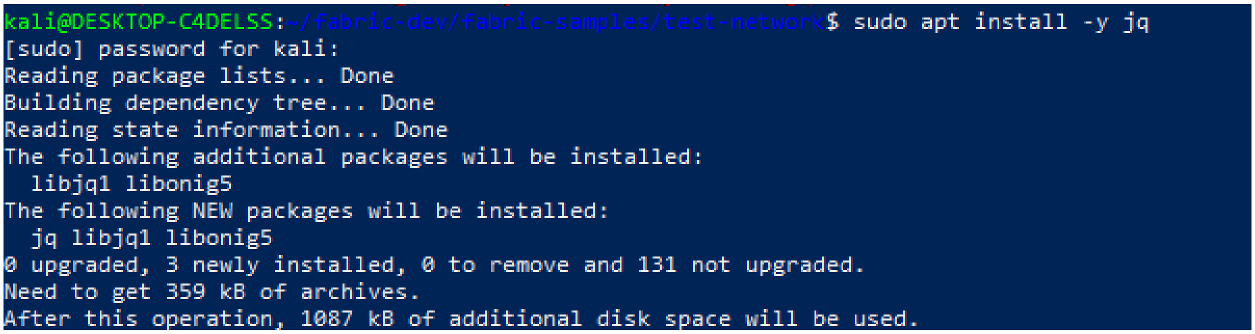
```
kali@DESKTOP-C4DELSS:~/fabric-dev$ cd fabric-samples
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples$ echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.bashrc
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples$ source ~/.bashrc
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples$ peer version
peer:
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger

kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples$ orderer version
orderer:
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64

kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples$ cd test-network
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ ./network.sh up
Using docker and docker compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from 'crypto
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
/home/kali/fabric-dev/fabric-samples/test-network/../bin/cryptogen
```

**Step 7:** Start test network

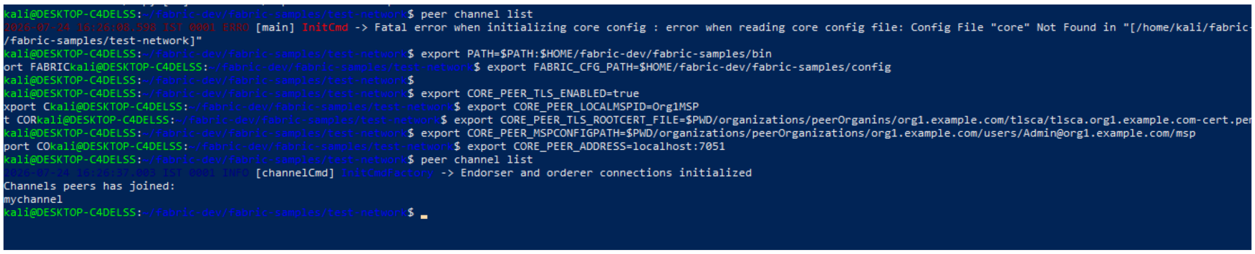
**Step 8:** Join peer organisation



```
                                                            vigorous_herschel
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ ./network.sh createChannel -c mychannel
Using docker and docker compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database '
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] up 3/3
 Container peer0.org2.example.com Running
ng                                                                                              0.0s  Co
                                                                          0.0sCONTAINER ID   IMAGE

9f8e4c12278   hyperledger/fabric-peer:latest      "peer node start"   38 seconds ago   Up 34 seconds
4->9444/tcp                                                 peer0.org1.example.com
6b21df9354e   hyperledger/fabric-peer:latest      "peer node start"   38 seconds ago   Up 34 seconds
, [::]:9445->9445/tcp                                       peer0.org2.example.com
544889fcb0d   hyperledger/fabric-orderer:latest   "orderer"           38 seconds ago   Up 34 seconds
3->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp   orderer.example.com
fcb8802a258b   hello-world                        "/hello"            24 minutes ago   Exited (0) 24 minutes ago
                                                            vigorous_herschel
Using docker and docker compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/kali/fabric-dev/fabric-samples/test-network/../bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-07-24 16:21:31.035 IST 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2026-07-24 16:21:31.056 IST 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type:
```

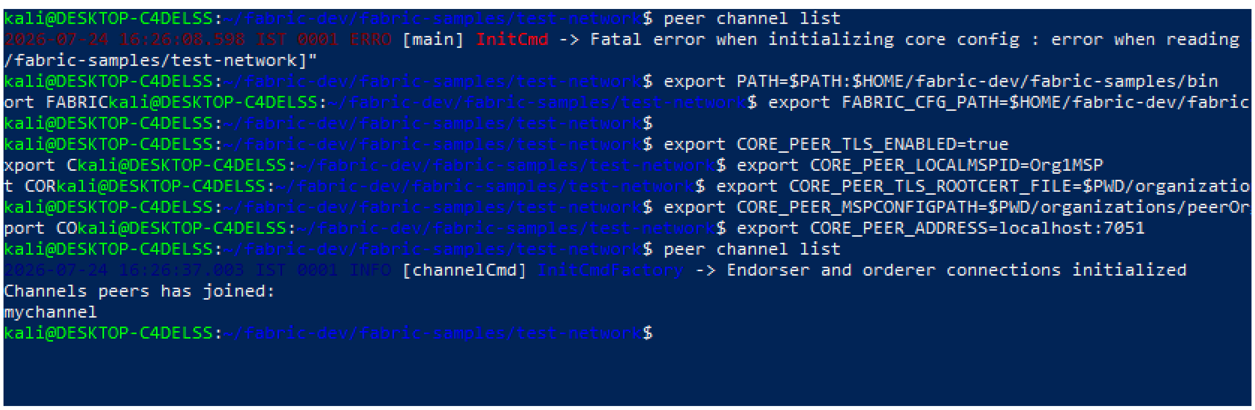


```
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ sudo apt install -y jq
[sudo] password for kali:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libjq1 libonig5
The following NEW packages will be installed:
  jq libjq1 libonig5
0 upgraded, 3 newly installed, 0 to remove and 131 not upgraded.
Need to get 359 kB of archives.
After this operation, 1087 kB of additional disk space will be used.
```



```
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ peer channel list
2026-07-24 16:26:08.598 IST 0001 ERRO [main] InitCmd -> Fatal error when initializing core config : error when reading core config file: Config File "core" Not Found in "[/home/kali/fabric-
/fabric-samples/test-network]"
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin
ort FABRICkali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-dev/fabric-samples/config
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_TLS_ENABLED=true
xport Ckali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_LOCALMSPID=Org1MSP
t CORkali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_TLS_ROOTCERT_FILE=$PWD/organizations/peerOrganins/org1.example.com/tlsca/tlsca.org1.example.com-cert.pe
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_MSPCONFIGPATH=$PWD/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
port COkali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_ADDRESS=localhost:7051
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ peer channel list
2026-07-24 16:26:37.003 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Channels peers has joined:
mychannel
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$
```

**Step 9:** Verify channel



```
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ peer channel list
2026-07-24 16:26:08.598 IST 0001 ERRO [main] InitCmd -> Fatal error when initializing core config : error when reading
/fabric-samples/test-network]"
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin
ort FABRICkali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-dev/fabric
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_TLS_ENABLED=true
xport Ckali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_LOCALMSPID=Org1MSP
t CORkali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_TLS_ROOTCERT_FILE=$PWD/organizatio
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_MSPCONFIGPATH=$PWD/organizations/peerOr
port COkali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ export CORE_PEER_ADDRESS=localhost:7051
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$ peer channel list
2026-07-24 16:26:37.003 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Channels peers has joined:
mychannel
kali@DESKTOP-C4DELSS:~/fabric-dev/fabric-samples/test-network$
```

**Result**

The Hyperledger Fabric development environment was successfully installed and configured. The required Fabric binaries and Docker images were installed, the test network was created successfully, the mychannel application channel was generated, both peer organizations joined the channel, and the anchor peers were configured successfully. The development environment is now ready for deploying chaincode (smart contracts) and developing blockchain applications.